TUTORIAL 5

IR Drops

SUBMITTED BY

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1 Goals

The goal of this lab to learn how power is delivered inside a chip and why voltage drop an important issue in modern IC design. When current flows through metal wires, some voltage is lost because of resistance. If this drop becomes too high, the circuit can slow down, give wrong results, or become less reliable. It causes timing violations, functional errors and reduce reliability.

Additionally, the lab also helps to learn how to run static and dynamic IR-drop analysis, how to find problem areas, and how to understand the overall power integrity of the chip.

1.1 Prepare tool

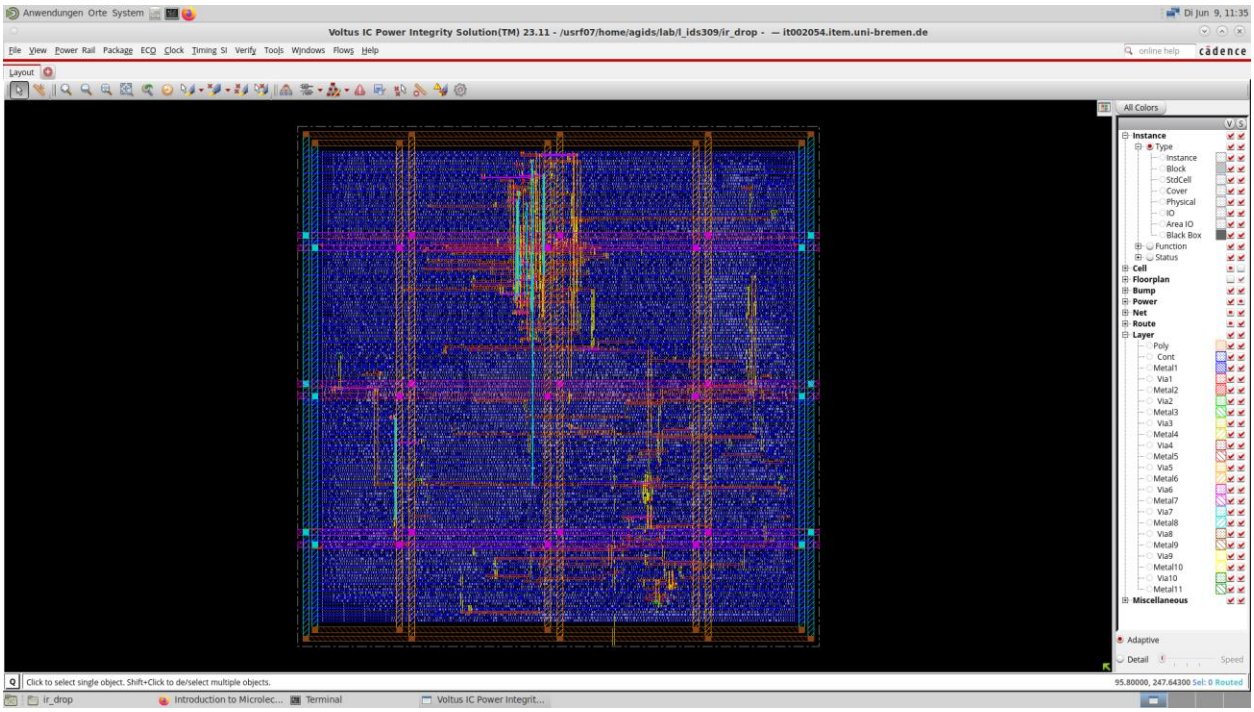
Before starting the design, we set up the folder structure and prepared the tool licenses needed for the IR-drop analysis.

2 Static Power Analysis

Static power is the power a chip uses even when it is not switching or doing any work. This power mainly comes from leakage currents inside the transistors.

Static power analysis in Voltus checks how much power the design uses when nothing is switching. Voltus estimates this leakage to help understand the chip's power behavior even when it is idle.

To analyze the power, we first loaded the design into Cadence Voltus using the LEF files, the MMMC view file, the Verilog file, and the DEF file.



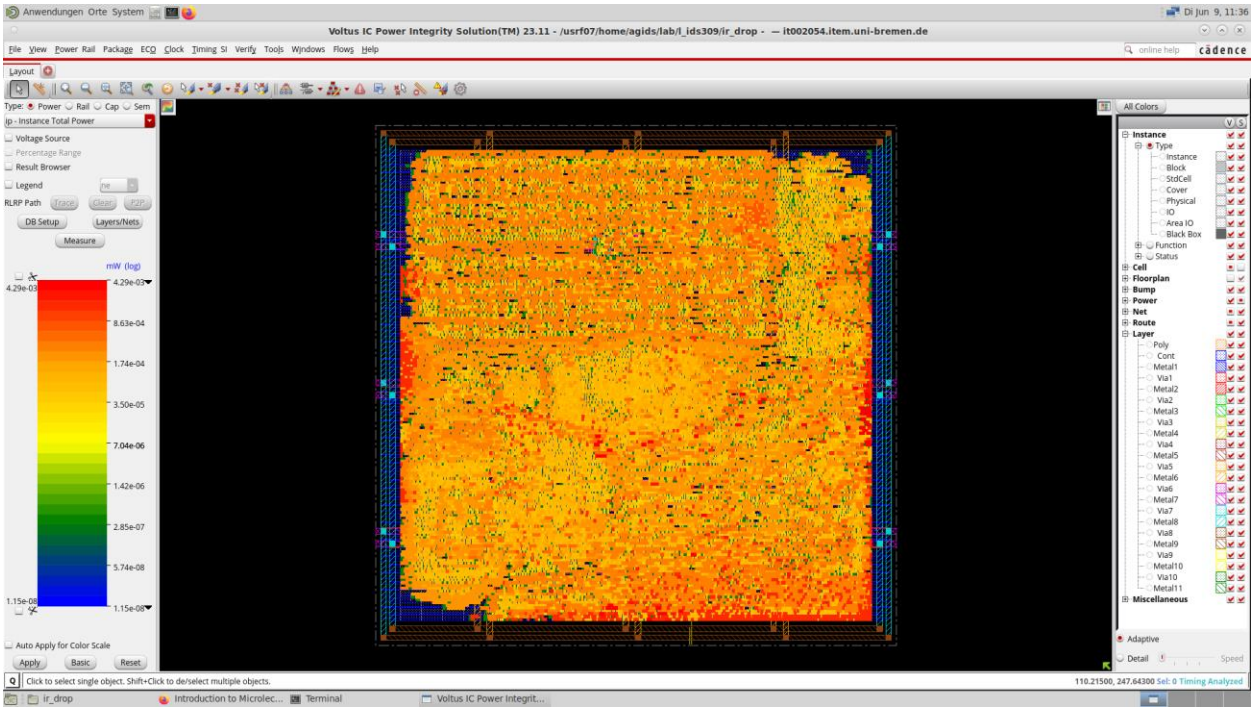
One of the most important input files required for power analysis is the SPEF (Standard Parasitic Exchange Format) file. A SPEF file contains the parasitic resistance and parasitic capacitance of

the interconnections in the chip after routing. It is generated by the physical design tool and is used to perform accurate timing, power, and signal integrity analysis.

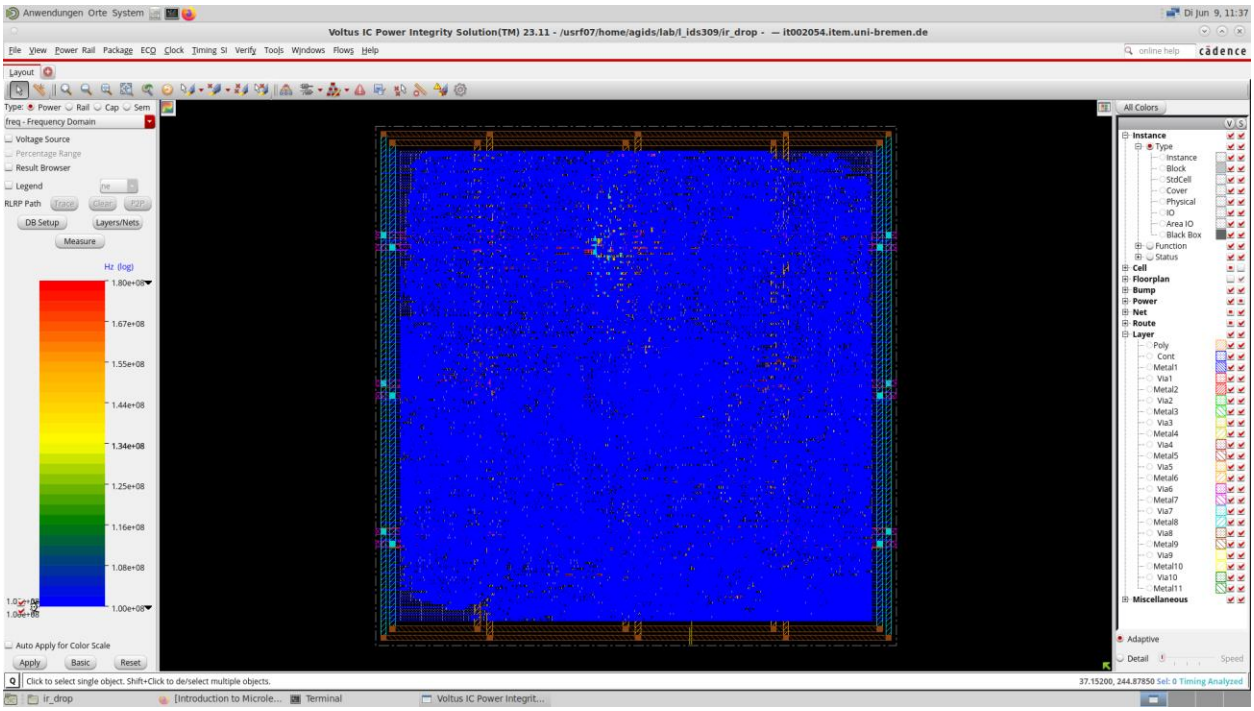
In this experiment, we loaded the SPEF file for the worst-case (RC_wc_125) analysis. After loading the design, we configured the power analysis mode by setting the leakage power view, dynamic power view, method, and enabling the writing of static current data. A binary database staticPower.db was also created to store the analysis results.

Next, we assigned switching activity to the rst input port and to the output pins of selected cells. After setting the activity values, we propagated the switching activity throughout the design so that Voltus could estimate the switching behavior of the entire circuit. We also defined the default switching activity, input activity, clock period, and clock gate output ratio for the remaining signals.

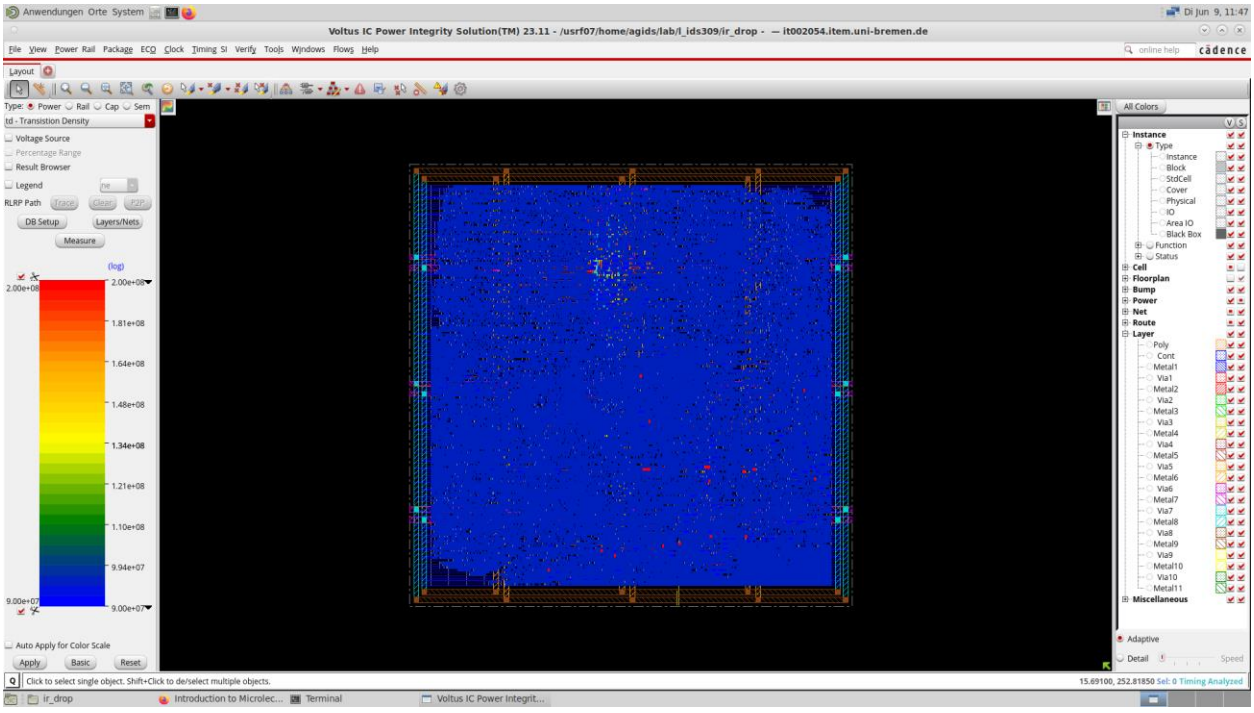
After completing the setup, we specified the output directory for storing the analysis results and ran the static power analysis. The tool generated a static power report static.rpt and a static power database staticPower.db. Finally, we loaded the generated power database and displayed the Instance Total Power plot, which shows the power consumption of different blocks in the design. This plot helps identify the blocks that consume the highest amount of power. The following plot shows the filtered instance power, highlighting specific instances.



Next, the following plot is the Frequency plot, which shows the operating frequency of different instances in the design.



Lastly, the Transition Density plot, which shows how often the signals switch during operation.



3 Static Rail Analysis

Static Rail Analysis is used to check the power delivery network of the chip by measuring the IR drop (voltage drop) across the power rails under static conditions. The purpose of this analysis is to make sure that every part of the chip receives enough supply voltage for correct operation.

First, we configured Voltus for static rail analysis by selecting the static analysis method, the worst-case analysis view (func_wc), and loading the required power grid library files. We also set the analysis accuracy and operating temperature to 125°C.

Next, we defined the power net **(**VDD**)** and ground net **(**VSS**)** by assigning their voltage values and threshold limits. These threshold values are used to detect IR drop violations if the supply voltage becomes too low or too high.

Then, we loaded the locations of the power and ground pads from the VDD.pp and VSS.pp files. These files contain the coordinates of the power pads and are required to model how current flows through the power grid.

In the next step, we loaded the average current data (.ptiavg files) that was generated during the static power analysis. Using this current information, Voltus performed the static rail analysis and stored the results in the staticRailResults directory. The reports are shown below:

```
Results for ir (linear filters):
================================

*
* Data filtering results for IR drop:
        Minimum, Average, Maximum IR Drop: 0.898845V, 0.899832V, 0.9V
        Range 1(0.898755V  - 0.898989V ):      105 (0.19%)
        Range 2(0.898989V  - 0.899134V ):       64 (0.11%)
        Range 3(0.899134V  - 0.899278V ):      243 (0.43%)
        Range 4(0.899278V  - 0.899423V ):      242 (0.43%)
        Range 5(0.899423V  - 0.899567V ):     1015 (1.81%)
        Range 6(0.899567V  - 0.899711V ):     2372 (4.23%)
        Range 7(0.899711V  - 0.899856V ):    24254 (43.28%)
        Range 8(0.899856V  -  0.90009V ):    27739 (49.50%)
```

When the global_activity is chsnged to a smaller value of 0.2, the changes are showing below:

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Results for ir (linear filters):
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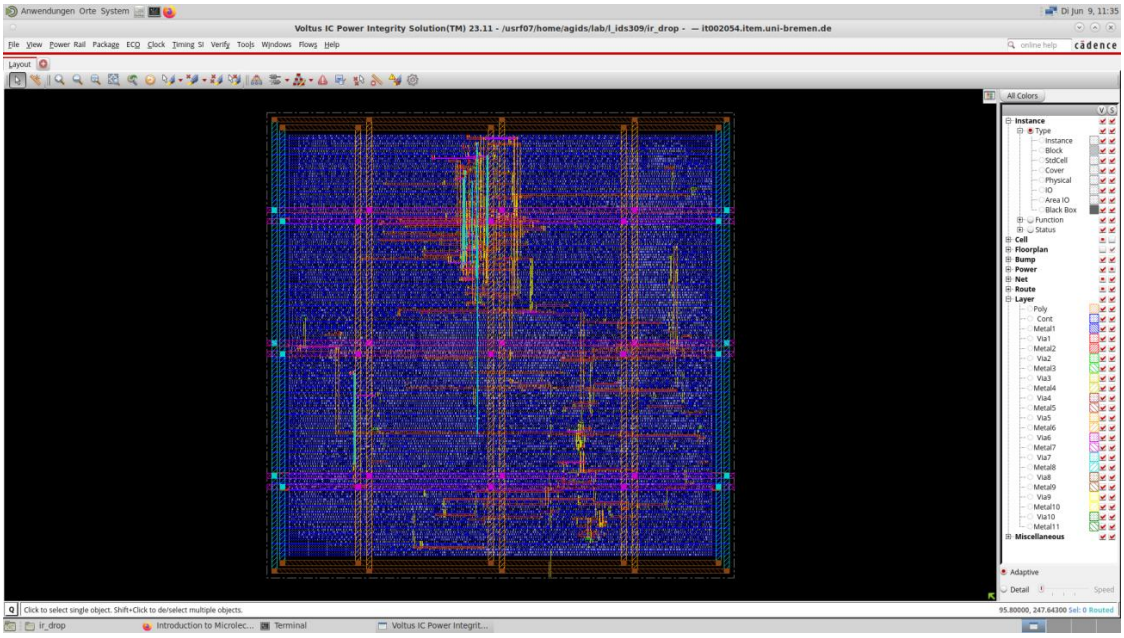
*
* Data filtering results for IR drop:
        Minimum, Average, Maximum IR Drop: 0.899746V, 0.89995V, 0.9V
        Range 1(0.899656V  - 0.899778V ):      110 (0.20%)
        Range 2(0.899778V  - 0.899809V ):       67 (0.12%)
        Range 3(0.899809V  - 0.899841V ):      330 (0.59%)
        Range 4(0.899841V  - 0.899873V ):      616 (1.10%)
        Range 5(0.899873V  - 0.899905V ):     1880 (3.36%)
        Range 6(0.899905V  - 0.899936V ):     7231 (12.90%)
        Range 7(0.899936V  - 0.899968V ):    34415 (61.42%)
        Range 8(0.899968V  -  0.90009V ):    11385 (20.32%)
```

The generated results showed the distribution of IR drop across the chip. We observed that the IR drop depends on the switching activity and power consumption of the design. Finally, we reduced the global switching activity from 0.95 to 0.1 and repeated the analysis. The new results showed a lower IR drop because the circuit consumed less power when the switching activity was reduced.

4 Dynamic Power Analysis

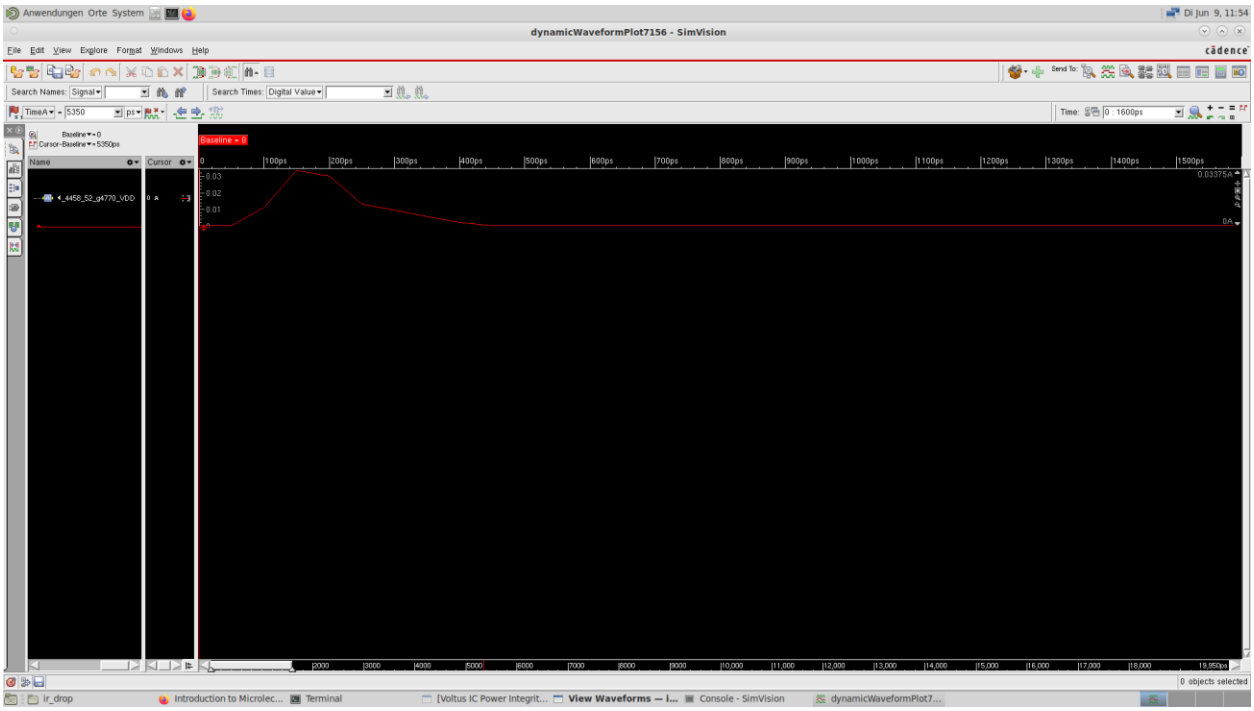
Dynamic power is the power consumed when the device is active, when signals are changing values (by switching logic states). Primary source of dynamic power consumption is switching power. Dynamic power depends on switching frequency, transition, output load and cell internal power.

First, we started Voltus and loaded all the required design files, including the LEF, Verilog, DEF, timing view, and SPEF files. These files provide the physical layout, circuit connections, timing information, and parasitic data required for accurate dynamic power analysis. This shows the following layout

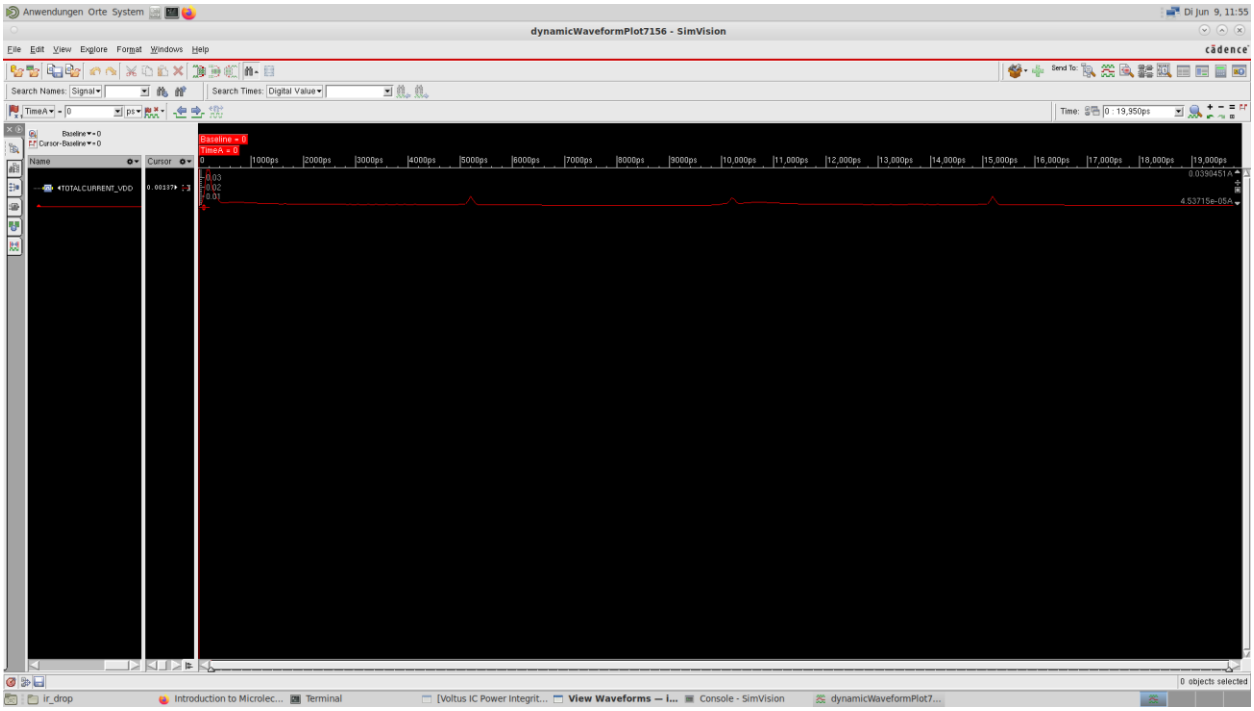


Next, we configured Voltus for dynamic vectorless power analysis. We selected the worst-case analysis view (func_wc), enabled the generation of static current data, and created a binary database (dynPower.db) to store the analysis results.

After that, we defined a current waveform for the selected instance (alu_sra_4458_52_g4770). The waveform specifies how the power changes with time during the simulation.



We also displayed the total current waveform of the entire VDD power net, which shows the overall current consumption of the complete design during operation. These waveforms helped us understand the dynamic power behavior of the chip and identify periods of high current consumption.



5 Dynamic Rail Analysis

Dynamic rail analysis is used to check the power integrity of the chip under real operating conditions. It analyzes the voltage drop (IR drop) in the power grid while the signals are switching and the current changes with time. This helps identify voltage fluctuations that may affect the performance of the chip.

First, we configured Voltus for dynamic rail analysis by selecting the dynamic analysis method and the worst-case analysis view (func_wc). We also enabled voltage waveform generation and loaded the required power grid library files. The analysis was performed at a temperature of 125°C.

Next, we defined the VDD and VSS power nets by assigning their voltage values , 0.9 ,and threshold limits, 0.05. Then, we created the rail analysis domain, PD_default , for the power and ground networks.

After that, we loaded the locations of the power and ground pads from the VDD.pp and VSS.pp files. These files contain the coordinates of the power pads and are required to accurately model the power distribution network.

Before running the analysis, we reset the previous power data and simulation settings and set the simulation time resolution to 50 ps. Then, we loaded the dynamic current waveform data for VDD and VSS.

After completing the setup, we ran the dynamic rail analysis. The results were stored in the dynVecLessRailResults directory, and a power report (dynrail.rpt) was generated.

Finally, we displayed the voltage waveform of the selected instance (alu_sra_4458_52_g4770). The waveform showed how the supply voltage changed over time because of the varying current demand during switching activity.

